

Improving Neural Network Performance

1. Vanishing Gradients

- Activation Functions
- Weight Initialization

2. Overfitting

- Reduce Complexity/Increase Data
- Dropout Layers
- Regularization (L1 & L2)
- Early Stopping

3. Normalization

- Normalizing inputs
- Batch Normalization
- Normalizing Activations

4. Gradient Checking and Clipping

5. Optimizers -

- Momentum
- Adagrad
- RMSprop
- Adam

6. Learning rate scheduling

7. Hyperparameter Tuning

- No. of hidden layers
- nodes/layer
- Batch size